

Suggested Solutions to 2016 'A' Level H2 Chemistry Paper 1 (9647/01)

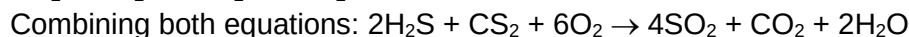
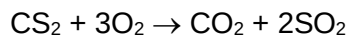
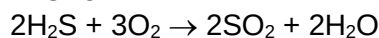
1 Answer: B

$$\text{Amount of Q} = 1.0 \text{ g} \div x \text{ g mol}^{-1} = 1/x \text{ mol}$$

$$\text{Number of molecules} = L \text{ mol}^{-1} \times (1/x) \text{ mol} = L/x$$

$$\text{Number of atoms} = 2(L/x) = 2L/x$$

2 Answer: D



$$\text{Mole ratio of SO}_2 : \text{CO}_2 = 4 : 1$$

3 Answer: C

$$A_r \text{ for Cu} = [65(63) + 29(65)] \div (65 + 29) = 63.6$$

4 Answer: D

$^{36}\text{S}^{2-}$: 16 protons, 20 neutrons, 18 electrons

$^{37}\text{Cl}^-$: 17 protons, 20 neutrons, 18 electrons

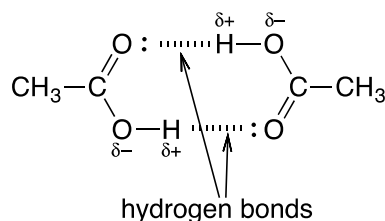
Option A: The nucleon numbers for S and Cl are 36 and 37 respectively.

Option B: Both ions have an outer electronic configuration of $3s^23p^6$.

Option C: Both ions have fewer electrons than neutrons.

Option D: Both ions have 20 neutrons in their nuclei.

5 Answer: A



6 Answer: C

There are three possible structures.

① $\text{C}^1-\text{C}^2\equiv\text{N}$ (C-1 has 1 lone pair and 1 unpaired electron; N has a lone pair)

② $\text{C}^1=\text{C}^2=\text{N}$ (C-1 has 1 lone pair; N has 1 lone pair and 1 unpaired electron)

③ $\text{C}^1\equiv\text{C}^2-\text{N}$ (C-1 has 1 unpaired electron; N has 2 lone pairs)

All three possible structures have 2 lone pairs of electrons and 1 unpaired electron.

7 Answer: C

By conservation of mass, mass of liquid = mass of vapour

$$pV = nRT$$

$$(101 \times 10^3)[(78 - 2) \times 10^{-6}] = (0.293/M)(8.31)(97 + 273)$$

$$\text{Molar mass, } M = 117.4 \text{ g mol}^{-1}$$

$$M_r \approx 117$$

8 Answer: D

Lattice energy is the enthalpy change when one mole of ionic compound is formed from its constituent gaseous ions under standard conditions.

Note: Lithium fluoride is a solid under standard conditions.

9 Answer: B

$$\Delta H_f \text{ of KCl} = 90 + \frac{1}{2}(242) + 418 + (-355) + (-710) = -436 \text{ kJ mol}^{-1}$$

- 10 Answer: D
 $\Delta G = \Delta H - T\Delta S$
- When ΔG is more negative at a higher temperature, ΔS must be positive.
OR
 1 mol of gas + 1 mol of solid $\rightarrow$ 2 mol of gases $\Rightarrow \Delta S > 0$ because a gas has greater entropy than a solid
- At a lower temperature, when “ $-T\Delta S$ ” is a small negative value, ΔG is positive. This implies that ΔH is positive.
OR
 $+78000 = \Delta H - 378\Delta S$
 Since $\Delta S > 0$, $-378\Delta S < 0$. Thus, ΔH must be positive.
- 11 Answer: C
 $E^\ominus_{\text{cell}} = E^\ominus(\text{Ag}^+/\text{Ag}) - E^\ominus(\text{Fe}^{3+}/\text{Fe}^{2+}) = (+0.80) - (+0.77) = +0.03 \text{ V}$
 To obtain a cell potential of 0.00 V, $E^\ominus(\text{Ag}^+/\text{Ag})$ needs to become less positive or $E^\ominus(\text{Fe}^{3+}/\text{Fe}^{2+})$ needs to become more positive, or both electrode potentials have to be adjusted to the same value.
 Option A: Increase in $[\text{Ag}^+]$ will make $E^\ominus(\text{Ag}^+/\text{Ag}) > +0.80 \text{ V}$
 Option B: Increase in $[\text{Fe}^{2+}]$ will make $E^\ominus(\text{Fe}^{3+}/\text{Fe}^{2+}) < +0.77 \text{ V}$
 Option C: Increase in $[\text{Fe}^{3+}]$ will make $E^\ominus(\text{Fe}^{3+}/\text{Fe}^{2+}) > +0.77 \text{ V}$ (can be adjusted to +0.80 V)
 Option D: Increase in the surface of the electrode does not change the electrode potential.
- 12 Answer: A
 Anode reaction: $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$
 Amount of electricity passed = $8 \text{ C s}^{-1} \times (100 \times 60) \text{ s} = 48000 \text{ C}$
 When $4 \times 96500 \text{ C}$ are passed, 1 mol of O_2 is liberated.
 When 48000 C are passed, $48000 / (4 \times 96500) = 0.1244 \text{ mol}$ of O_2 is liberated.
 Volume of O_2 liberated at s.t.p. = $0.1244 \times 22.7 = 2.8 \text{ dm}^3$ (1 d.p.)
- 13 Answer: C
 X is a saturated solution of ZnF_2 .
 $\text{ZnF}_2(\text{s}) \rightleftharpoons \text{Zn}^{2+}(\text{aq}) + 2\text{F}^-(\text{aq}) \quad K_{\text{sp}} = [\text{Zn}^{2+}][\text{F}^-]^2 = 3.2 \times 10^{-2} \text{ mol}^3 \text{ dm}^{-9}$
 Let $[\text{Zn}^{2+}]$ be $y \text{ mol dm}^{-3}$ and $[\text{F}^-]$ be $2y \text{ mol dm}^{-3}$.
 $y(2y)^2 = 3.2 \times 10^{-2}$
 $[\text{F}^-] = 2y = 4 \times 10^{-1} \text{ mol dm}^{-3}$
 When BaF_2 just precipitates, $[\text{Ba}^{2+}][\text{F}^-]^2 = 1.6 \times 10^{-7} \text{ mol}^3 \text{ dm}^{-9}$
 Since $[\text{F}^-] = 4 \times 10^{-1} \text{ mol dm}^{-3}$, $[\text{Ba}^{2+}] = 1.6 \times 10^{-7} \div (4 \times 10^{-1})^2 = 1 \times 10^{-6} \text{ mol dm}^{-3}$
- 14 Answer: B
 $\text{Al}(\text{H}_2\text{O})_6^{3+}(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{Al}(\text{OH})(\text{H}_2\text{O})_5^{2+}(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$
 $[\text{H}^+] = \sqrt{(1.0 \times 10^{-5} \times 0.1)} = 1.0 \times 10^{-3} \text{ mol dm}^{-3}$
 $\text{pH} = -\log(1.0 \times 10^{-3}) = 3.0$
- 15 Answer: D
 $y = k(a)(a)^2(a)^2$
 Hence, $k = y/a^5$
 $\text{rate} = (y/a^5)(a/2)(2a)^2(3a)^2 = 18y$
- 16 Answer: A
 Magnesium oxide, though having a giant ionic lattice, has less covalent character than aluminium oxide. Phosphorus pentoxide and silicon dioxide are predominantly covalent, so they do not exist as giant ionic lattices.
- 17 Answer: A
 $\text{Mg}(\text{NO}_3)_2(\text{s}) \rightarrow \text{MgO}(\text{s}) + 2\text{NO}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g})$

Amount of $\text{Mg}(\text{NO}_3)_2 = 10.4 / 148.3 = 0.07013 \text{ mol}$
 Amount of O_2 (neutral gas) = $\frac{1}{2} (0.07013) = 0.03507 \text{ mol}$
 Mass of O_2 (neutral gas) = $0.03507 \times 32.0 = 1.12 \text{ g}$

- 18 Answer: A
 Option B: Magnesium (m.p. 650°C) has a higher melting point than sulfur (m.p. 115.2°C) because the energy needed to overcome the metallic bonds in the giant metallic lattice of magnesium is greater than that needed to overcome the id-id interactions between S_8 molecules in the simple molecular lattice of sulfur.
 Option C: Magnesium hydroxide is only sparingly soluble in water.
 Option D: Magnesium reacts slowly with cold water.
- 19 Answer: D
 White silver chloride formed dissolves in concentrated aqueous ammonia to give a colourless solution.
 Cream silver bromide formed dissolves in concentrated aqueous ammonia to give a colourless solution.
 Yellow silver iodide formed is insoluble in concentrated aqueous ammonia.
- 20 Answer: D
 Element Y is vanadium.
- 21 Answer: B
 $2\text{VO}_2^+ + \text{SO}_2 \rightarrow 2\text{VO}^{2+} + \text{SO}_4^{2-} \quad E^\ominus_{\text{cell}} = 1.00 - 0.17 = +0.83 \text{ V (feasible)}$
 $2\text{VO}^{2+} + \text{SO}_2 \rightarrow 2\text{V}^{3+} + \text{SO}_4^{2-} \quad E^\ominus_{\text{cell}} = 0.34 - 0.17 = +0.17 \text{ V (feasible)}$
 $2\text{V}^{3+} + \text{SO}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{V}^{2+} + \text{SO}_4^{2-} + 4\text{H}^+ \quad E^\ominus_{\text{cell}} = -0.26 - 0.17 = -0.43 \text{ V (not feasible)}$
- 22 Answer: B
 NH_3 ligand has no charge, while Cl^- ligand has a charge of $1-$.
 Let the number of NH_3 ligands be $6 - n$ and the number of Cl^- ligands be n .
 Charge on cation in platinum(IV) compound = $(+4) + (6 - n)(0) + n(-1) = 2$
 Solving, $n = 2$, i.e. there are 2 Cl^- ligands. Hence, there are 4 NH_3 ligands.
 $\text{PtCl}_4 + 4\text{NH}_3 \rightarrow [\text{Pt}(\text{NH}_3)_4\text{Cl}_2]^{2+} + 2\text{Cl}^-$
 Option A: The cation, $[\text{Pt}(\text{NH}_3)_3\text{Cl}_3]^+$, has a $1+$ charge.
 Option C: The oxidation state of platinum in $[\text{Pt}(\text{NH}_3)_6]^{2+}$ is $+2$.
 Option D: The cation, $[\text{Pt}(\text{NH}_3)_6]^{4+}$, has a $4+$ charge.
- 23 Answer: B
 There are 2 π bonds found in the $\text{C}\equiv\text{C}$ bond.
 Both C atoms in the $\text{C}\equiv\text{C}$ bond are sp hybridised.
 The remaining C in CH_3 is sp^3 hybridised.
- 24 Answer: A
 A termination step involves the collision of 2 free radicals. Hence, option B is incorrect.
 An $\text{H}\bullet$ free radical is not formed, so options C and D are incorrect.
- 25 Answer: D
 The rate-determining step of this $\text{S}_{\text{N}}1$ mechanism involves the breaking of the $\text{C}-\text{Br}$ or $\text{C}-\text{Cl}$ bond.
 Since the $\text{C}-\text{Br}$ bond is weaker, reaction 1 is faster.
- 26 Answer: C
 Na reacts with the phenolic group (ROH) in compound C.
 $\text{ROH} + \text{Na} \rightarrow \text{RO}^-\text{Na}^+ + \frac{1}{2}\text{H}_2$
- 27 Answer: A
 Option B shows a compound which does not have the empirical formula CH_2O .
 Option C and option D show compounds which do not produce yellow CHI_3 in the iodoform test because they are neither methyl alcohols nor methyl ketones. In addition, neither one is ethanal.

28 Answer: B
When 3-bromo-4-hydroxycinnamic acid is dissolved in water, its phenolic and carboxylic acid groups will readily ionise to release H^+ . The resulting conjugate bases may remove a D^+ ion from D_2O .
 $\text{ROH} + \text{D}_2\text{O} \rightarrow \text{ROD} + \text{HDO}$ and $\text{RCOOH} + \text{D}_2\text{O} \rightarrow \text{RCOOD} + \text{HDO}$

29 Answer: B

- As X is a carbonyl compound which can be reduced by NaBH_4 , it could be either a ketone or an aldehyde.
- As X does not react with alkaline aqueous iodine or with Tollens' reagent, it is neither a methyl ketone (RCOCH_3) nor an aldehyde.

Hence, X must be a ketone which does not have the $-\text{COCH}_3$ functional group.
The reduction of X will give a secondary alcohol which does not have the $-\text{CH}(\text{OH})\text{CH}_3$ functional group.
Option A shows a tertiary alcohol.
Option C shows a secondary alcohol with a $-\text{CH}(\text{OH})\text{CH}_3$ functional group.
Option D shows a primary alcohol.

30 Answer: B
Option A and option D show the hydrolysis of esters, a relatively slow reaction which usually requires heating and the use of a catalyst such as dilute H_2SO_4 .
Option B shows the hydrolysis of an acyl chloride, which occurs readily at room temperature because the electron-rich O atom in water can readily attack the highly electron-deficient, sp^2 C atom in the acyl group.
Option C shows the hydrolysis of an amide, a reaction which requires prolonged heating in an acid.

31 Answer: B (1 and 2 only)

Option 1

Isotope	^4He	^{12}C	^{24}Mg
No. of protons	2	6	12
No. of neutrons	2	6	12

Option 2

Isotope	^{14}N	^{20}Ne	^{30}P
No. of protons	7	10	15
No. of neutrons	7	10	15

Option 3

Isotope	^{28}Si	^{34}S	^{40}Ca
No. of protons	14	16	20
No. of neutrons	14	18	20

32 Answer: A (1, 2 and 3)

Option 1

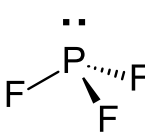
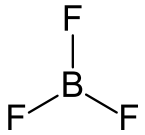
Molecule	SO_2	CO_2
Structure	$\begin{array}{c} \cdot\cdot \\ \text{O}=\text{S}=\text{O} \end{array}$	$\text{O}=\text{C}=\text{O}$
Net dipole	Yes	No

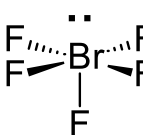
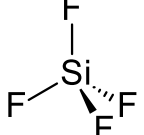
Option 2

Molecule	PF_3	BF_3
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Option 3

Molecule	BrF_5	SiF_4
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Structure		
Net dipole	Yes	No

Structure		
Net dipole	Yes	No

33 Answer: C (2 and 3 only)

Brønsted-Lowry acid: H^+ donor

Brønsted-Lowry base: H^+ acceptor

Option 1: NH_3 is a nucleophile, while CH_3Cl is an electrophile.

Option 2: CH_3OH accepted an H^+ ion, while HClO_4 donated an H^+ ion.

Option 3: HNO_3 accepted an H^+ ion, while H_2SO_4 donated an H^+ ion.

34 Answer: D (1 only)

The ethoxide ion, $\text{C}_2\text{H}_5\text{O}^-$, is a nucleophile and it attacks the primary iodoalkane, CH_3I , in a nucleophilic substitution reaction.

From experiments 1 and 3, when $[\text{C}_2\text{H}_5\text{ONa}]$ is doubled and $[\text{CH}_3\text{I}]$ is kept constant, the relative initial rate of reaction is doubled. Hence, the reaction is first order with respect to $[\text{C}_2\text{H}_5\text{ONa}]$. So, option 1 is correct.

From experiments 2 and 3, when $[\text{C}_2\text{H}_5\text{ONa}]$ is kept constant and $[\text{CH}_3\text{I}]$ is halved, the relative initial rate of reaction is halved. Hence, the reaction is first order with respect to $[\text{CH}_3\text{I}]$. So, option 3 is incorrect.

The overall order reaction is $1 + 1 = 2$. So, option 2 is incorrect.

35 Answer: C (2 and 3 only)

Brønsted-Lowry acid: H^+ donor

Option 1: BeCl_2 did not donate any H^+ ions, so it is not a Brønsted-Lowry acid.

Option 2: $\text{Be}(\text{H}_2\text{O})_4^{2+}$ donated 2 H^+ ions to 2 Cl^- ions, so $\text{Be}(\text{H}_2\text{O})_4^{2+}$ is a Brønsted-Lowry acid.

Option 3: $\text{Be}(\text{H}_2\text{O})_2(\text{OH})_2$ donated 2 H^+ ions to 2 OH^- ions, so $\text{Be}(\text{H}_2\text{O})_2(\text{OH})_2$ is a Brønsted-Lowry acid.

36 Answer: B (1 and 2 only)

oxidation no.

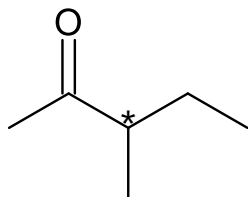
0
 Cl_2
reactant

-1 +5
 NaCl NaClO_3
products

37 Answer: C (2 and 3)

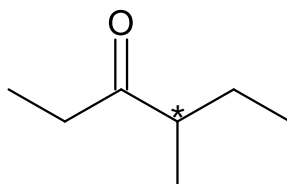
There is no such molecule with $x = 5$.

When $x = 6$



E.g. 3-methyl-pentan-2-one

When $x = 7$

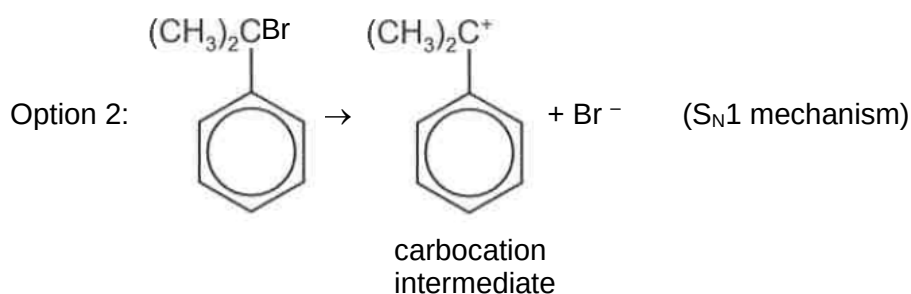


E.g. 4-methyl-hexan-3-one

38 Answer: B (1 and 2 only)

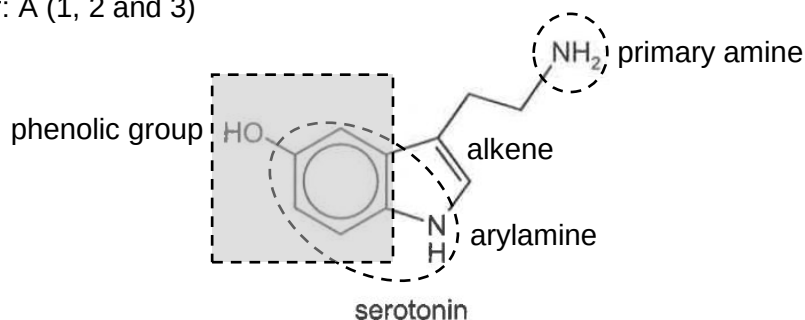
Option 1: $\text{OH}^- + \text{CH}_3\text{Cl} \rightarrow [\text{HO} \cdots \text{CH}_3 \cdots \text{Cl}]^-$ ($\text{S}_\text{N}2$ mechanism)

transition state



Option 3: This cationic species is not formed in a nucleophilic substitution reaction.

39 Answer: A (1, 2 and 3)



Option 1: $\text{CH}_3\text{COC}l$ reacts with the phenolic group to give an ester. It also reacts with the primary amine and the arylamine to give amides.

Option 2: HCl neutralises the amines to give salts. It also undergoes addition with the alkene.

Option 3: NaOH neutralises the acidic phenolic group to give a phenoxide salt.

40 Answer: C (2 and 3 only)

The polypeptide given is glycine–cysteine–lysine–glycine–lysine (N terminus on the left and C-terminus on the right).

Action of trypsin: glycine–cysteine–lysine–|–glycine–lysine

Possible fragments: glycine–cysteine–lysine, glycine–lysine